SIMPLEST FORM

Rationalizing a monomial denominator

A radical in the denominator violates simplest form. One multiplication removes it without changing the value.

BEFORE

Radical in the denominator

$$\frac{5}{\sqrt{3}}$$

Not in simplest form

RATIONALIZE

Multiply by $\frac{\sqrt{3}}{\sqrt{3}}$

CUBE ROOTS

AFTER

Rational denominator

$$\frac{5\sqrt{3}}{3}$$

Simplest form: the denominator is the rational number 3

WHAT CHANGED

The denominator: $\sqrt{3} \cdot \sqrt{3} = \sqrt{9} = 3$, a rational number.

WHAT STAYED

The value: the multiplier goes into both the numerator and the denominator, so only the form changes.